

- **Hadamard gate (H):** Creates superposition by turning a definite state (0 or 1) into a combination of both.
- **Pauli-X gate:** Similar to a classical NOT gate, it flips the state of a qubit.
- **CNOT gate (Controlled-NOT):** Entangles two qubits, flipping the second qubit if the first is in state 1.

Quantum circuits are formed by arranging these gates in sequence, enabling the execution of complex quantum algorithms. For example, a simple quantum circuit might take an input qubit, place it in superposition with a Hadamard gate, and then entangle it with another qubit using a CNOT gate.

Decoding quantum algorithms

Quantum algorithms are specially crafted procedures that exploit the unique properties of quantum computers to solve problems. While the mathematics behind them can be intricate, the basic idea is to use superposition and entanglement to explore many possible solutions at once.

A classic example is Grover's algorithm, which can search an unsorted database much faster than any classical algorithm. For instance, if you had to find a specific name in a phone book with 10 million entries, a classical computer might check, on average, 5 million entries before finding the right one. Grover's algorithm, running on a quantum computer, could locate the correct entry in just about 10,000 steps—a dramatic speed-up. Another well-known algorithm is Shor's algorithm, designed to factor large numbers efficiently. This has significant implications for cybersecurity, as much of modern encryption depends on the difficulty of this task for classical computers.

Beyond these, quantum algorithms are being developed for optimisation, machine learning, and simulation of physical systems—areas where quantum computers could eventually outperform their classical counterparts.

Example quantum algorithm: Grover's search algorithm

One of the most elegant and widely taught quantum algorithms is Grover's search algorithm, designed by Lov Grover in 1996. It demonstrates how quantum computing can offer a significant speedup for solving unstructured search problems—an area where classical computers struggle.

The problem: Imagine you are searching for a specific item in an unsorted database with N entries—for example, finding one name in a list of a million without any sorting or indexing.

- A classical computer must check each entry one by one, taking on average $N/2$ steps, or $O(N)$ time.
- Grover's algorithm, using quantum parallelism, can locate the correct entry in approximately $\sqrt{N}$ steps, an enormous improvement for large datasets.

Quantum principles at work:

Grover's algorithm leverages three core quantum mechanics principles.

- **Superposition:** Represents all possible database entries simultaneously.
- **Interference:** Amplifies the probability of the correct answer while diminishing incorrect ones.
- **Measurement:** Collapses the system to reveal the most probable solution—the desired item.

Step-by-step conceptual overview:

- **Step 1 – Initialisation:** The algorithm begins by placing all possible states (entries) into superposition. This means the quantum computer considers every entry at once rather than sequentially.
- **Step 2 – Oracle function:** An 'oracle' is a black-box function that can identify whether a particular state is the correct solution. When applied, it flips the sign of the correct state's amplitude without revealing which one it is.
- **Step 3 – Amplification (Grover iteration):** Through a process

called amplitude amplification, the algorithm increases the probability of the correct answer while reducing the likelihood of incorrect ones. It achieves this using interference, combining wave-like quantum states constructively for the correct answer and destructively for others.

- **Step 4 – Repetition:** The Grover iteration is repeated approximately $\sqrt{N}$ times. After each iteration, the probability of measuring the correct solution grows.
- **Step 5 – Measurement:** Finally, the system is measured, collapsing the quantum superposition into a definite state. The correct answer appears with high probability.

Classical vs quantum efficiency:

Here's the comparison:

Aspect	Classical search	Grover's quantum search
Search space size	(N)	(N)
Steps required	$(O(N))$	$(O(\sqrt{N}))$
Efficiency gain	Linear	Quadratic

While this may seem modest compared to exponential speedups (like in Shor's algorithm), the quadratic improvement becomes transformative as (N) grows.

Real-world implications: Grover's algorithm shows the quantum advantage in search and optimisation problems. Potential applications include:

- Database searching
- Pattern recognition
- Cryptanalysis (e.g., accelerating brute-force attacks)
- AI optimisation tasks

Though it doesn't make encryption instantly obsolete, Grover's algorithm emphasises the need for quantum-resistant cryptography, as it can halve the effective key length of symmetric encryption methods like AES.